

Automating FAA Flight Delay Data Collection

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Introduction

Goal: Automate fetching FAA flight delay reports from the FAA Website.

Key Steps:

- Replicate browser HTTP POST requests using Python.
- Extract headers, cookies, and payload from FAA's website.
- Save the response as an HTML report.

Tools Used:

- Python requests library.
- Browser Developer Tools (for inspecting requests).

Key Insights:

- **Automation:** Avoids manual data fetching.
- **Flexibility:** Modify `payloadDict` for other airports/dates.
- **Challenges:**
 - Session/cookie expiration.
 - Dynamic payload adjustments.

HTTP (Hypertext Transfer Protocol):

- Application-layer protocol for client-server communication
- Stateless by design (each request independent)
- Uses request-response model

Key Components

Headers	Metadata about request/response
Cookies	Client-side state management
Payload	Data sent in request body
Methods	GET (read) vs POST (write)

HTTP Headers

Purpose: Carry metadata about the HTTP transaction

Common Request Headers:

- User-Agent: Client software
- Content-Type: Data format (e.g., JSON)
- Authorization: Credentials

Common Response Headers:

- Set-Cookie: Server-side cookies
- Content-Length: Payload size
- Status: Response code (200 OK)

Purpose: Maintain state across stateless HTTP requests

- Small pieces of data stored client-side
- Sent with every request to the same domain
- Typical uses:
 - Session management (e.g., ASPSESSIONID)
 - Personalization
 - Tracking

Definition: The actual data sent in the HTTP request body

- Used with POST/PUT requests (GET appends to URL)
- Common formats:
 - Form data (`application/x-www-form-urlencoded`)
 - JSON (`application/json`)
 - XML

GET vs POST

GET

- Requests data from server
- Parameters in URL
- Bookmarkable/cacheable
- Idempotent (safe to repeat)

POST

- Submits data to server
- Parameters in request body
- Not bookmarkable
- Non-idempotent (may change state)



Federal Aviation
Administration

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[MyAccess Login](#)

[TFMSC Main](#) [Airport](#) [Distributed OPSNET](#)

[FAA Operations & Performance Data](#)

Traffic Flow Management System Counts > Airport View



My Reports ?

Please use option buttons above to specify your report criteria, and then click **Run**.

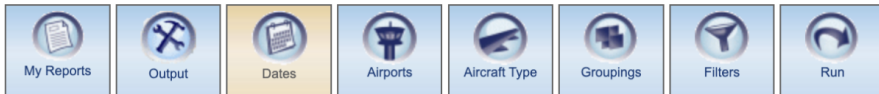


U.S. Department of Transportation

Federal Aviation Administration
800 Independence Avenue, SW
Washington, DC 20591
1-866-TELL-FAA (1-866-835-5322)

Select the dates to query:

Traffic Flow Management System Counts > Airport View



Select Dates ?

☒ Days

☐ Months

☐ Years

☐ Range

Sun	Mon	Tue	Wed	Thu	Fri	Sat
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30	31				

<< December >>
<<< 2024 >>>

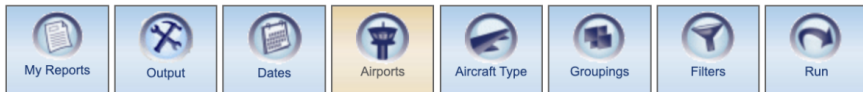
12/12/2024 ✖
12/13/2024 ✖
12/14/2024 ✖
12/15/2024 ✖
12/16/2024 ✖
12/17/2024 ✖
12/18/2024 ✖

Selected options:

Dates : 12/12/2024 12/13/2024
12/14/2024 12/15/2024 12/16/2024
12/17/2024 12/18/2024
Grouping : Date; Airport; User
Class; Weight Class; Physical
Class; Aircraft Type; Business Jet;
Regional Jet; Business Aviation;
Airplane Approach Category (AAC);
Airplane Design Group (ADG);
Taxiway Design Group (TDG); Flight
Type
Facilities : MCO
Display : Show Totals
Format : HTML

Type in the interested airport(s) ICAO ID(s)

Traffic Flow Management System Counts > Airport View



Select Airports ?

Enter an airport codes:

Go

☒ Quick Entry Mode

[ASPM 82](#) | [OPSNET 45](#) | [Core](#) | [Advanced Search Options](#)

☒ MCO - Orlando

[Remove All](#)

Selected options:

Dates : 12/12/2024 12/13/2024 12/14/2024
12/15/2024 12/16/2024 12/17/2024 12/18/2024
Grouping : Date; Airport; Flight Type; User Class;
Weight Class; Physical Class; Aircraft Type; Business
Jet; Regional Jet; Business Aviation; Airplane
Approach Category (AAC); Airplane Design Group
(ADG); Taxiway Design Group (TDG)
Facilities : MCO
Display : Show Totals
Format : HTML

Select which field you want to aggregate :

Traffic Flow Management System Counts > Airport View



Select Groupings ?

Available Fields



[Remove All](#)

Selected Fields



Selected options:

Dates : 12/12/2024 12/13/2024 12/14/2024
12/15/2024 12/16/2024 12/17/2024 12/18/2024
Grouping : Date; Airport; Flight Type; User Class;
Weight Class; Physical Class; Aircraft Type; Business
Jet; Regional Jet; Business Aviation; Airplane
Approach Category (AAC); Airplane Design Group
(ADG); Taxiway Design Group (TDG)
Facilities : DAB
Display : Show Totals
Format : HTML

Click the Run icon on the top right corner to see a result page like below.

TFMSC Report (Airport)

From 12/12/2024 To 12/18/2024 | Airport=MCO

#	Date	Airport	Flight Type	User Class	Weight Class	Physical Class	Aircraft Type
1	12/12/2024	MCO - Orlando	Foreign to US	Air Carrier	?	Jet	E295 - unk
2	12/12/2024	MCO - Orlando	Foreign to US	Air Carrier	Heavy Eqpt	Jet	A332 - Airt
3	12/12/2024	MCO - Orlando	Foreign to US	Air Carrier	Heavy Eqpt	Jet	A333 - Airt
Sub-Total for 5							
Sub-Total for V							
Sub-Total for C							
Sub-Total for No							
4	12/12/2024	MCO - Orlando	Foreign to US	Air Carrier	Heavy Eqpt	Jet	A339 - A3
5	12/12/2024	MCO - Orlando	Foreign to US	Air Carrier	Heavy Eqpt	Jet	B772 - Bo

- 1 What will happen if we refresh this page? What HTTP request (GET/POST) is sent to the FAA server?
- 2 Can we generate a similar request using the request library and obtain a similar response?

Step 1: Extract HTTP Headers

Browser: Developer Tool - Network tab's **Headers**.

Refresh Browser if **Headers** is not seen!!

The screenshot shows the Chrome DevTools Network tab. The top toolbar includes icons for Elements, Console, Sources, Network (selected), Performance, Memory, Application, and Security. Below the toolbar, there are checkboxes for 'Preserve log', 'Disable cache', and 'No throttling'. A filter bar shows 'Filter' and 'Invert' options, followed by a list of filter categories: All (selected), Fetch/XHR, Doc, CSS, JS, Font, Img, Media, Manifest, WS, Wasm, and Other. A timeline at the top shows a request starting at 200 ms and ending at 1,600 ms. The main panel displays the 'Headers' tab for the selected request 'tfms-server-x.asp'. The 'General' section shows the Request URL as 'https://aspm.faa.gov/tfms/sys/tfms-server-x.asp', Request Method as 'POST', Status Code as '200 OK', Remote Address as '96.7.198.167:443', and Referrer Policy as 'strict-origin-when-cross-origin'. The 'Response Headers' section is expanded, showing various headers including 'Access-Control-Allow-Headers', 'Access-Control-Allow-Methods', 'Access-Control-Allow-Origin', 'Cache-Control', 'Connection', 'Content-Type', 'Date', 'Expires', and 'Set-Cookie'.

Name	Value
Request URL:	https://aspm.faa.gov/tfms/sys/tfms-server-x.asp
Request Method:	POST
Status Code:	200 OK
Remote Address:	96.7.198.167:443
Referrer Policy:	strict-origin-when-cross-origin
Response Headers	
Access-Control-Allow-Headers:	Origin, X-Requested-With, Content-Type, Accept
Access-Control-Allow-Methods:	POST,GET,OPTIONS,PUT,DELETE
Access-Control-Allow-Origin:	https://aspm.faa.gov
Cache-Control:	private
Connection:	keep-alive
Connection:	Transfer-Encoding
Content-Type:	text/html
Date:	Sat, 01 Feb 2025 16:07:53 GMT
Expires:	Sat, 01 Feb 2025 16:07:53 GMT
Set-Cookie:	bm_sv=09033781AE5C60B79EC8B2DAFB405057~YAAQh8BMaGJDwJUUAQAAZMBEwhhCCC5eb9CXFiue1SoUjVwIDklOY0DgaPwXK/DOALrnpMRarag2U

Step 1: Extract HTTP Headers

header.

X Headers Payload Preview Response Initiator Timing Cookies	
▼ General	
Request URL:	https://aspm.faa.gov/tfms/sys/tfms-server-x.asp
Request Method:	POST
Status Code:	200 OK
Remote Address:	96.7.198.167:443
Referrer Policy:	strict-origin-when-cross-origin
▼ Response Headers ☐ Raw	
Access-Control-Allow-Headers:	Origin, X-Requested-With, Content-Type, Accept
Access-Control-Allow-Methods:	POST,GET,OPTIONS,PUT,DELETE
Access-Control-Allow-Origin:	https://aspm.faa.gov
Cache-Control:	private
Connection:	keep-alive
Connection:	Transfer-Encoding
Content-Type:	text/html
Date:	Sat, 01 Feb 2025 16:07:53 GMT
Expires:	Sat, 01 Feb 2025 16:07:53 GMT
Set-Cookie:	bm_sv=09033781AE5C60B79EC8B2DAF405057-YAAQh8BMGJDWJuUAQAAzMBEwhpCCC5eb9CXElue1SoUVwJDKLOY0DqaPwKK/DOALrnpMRarag2U1LnwM3tM44adyFN637ml+Dkc3VPYxqT4/mpPK9l0cqb1W6BI7lt+dKFTl+gILS MJxXTDK8so4lhvDtpu3k4+SmgJFO0naB5ADI/WT+QBEAx0LYwQebJFfouMF KQI4eWN9akMWgMDh/HpGMKPvWQHUUnDh6S50vOOBpt5hCbOB4ySo-1; Domain=.faa.gov; Path=/; Expires=Sat, 01 Feb 2025 18:03:27 GMT; Max-Age=6934; Secure
Strict-Transport-Security:	max-age=31536000; includeSubDomains
Transfer-Encoding:	chunked

▼ Request Headers ☐ Raw	
Accept:	text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng/*;q=0.8,application/signed-exchange;v=b3;q=0.7
Accept-Encoding:	gzip, deflate, br, zstd
Accept-Language:	en-US,en;q=0.9
Cache-Control:	max-age=0
Connection:	keep-alive
Content-Length:	1613
Content-Type:	application/x-www-form-urlencoded

X Headers Payload Preview Response Initiator Timing Cookies		
Form Data	view source	view URL-encoded
dstyle: d		
dfld: yyyymdd		
dlilst: 20241212, 20241213, 20241214, 20241215, 20241216, 20241217, 20241218		
fromdate:		
todate:		
lilst: ' MC0'		
elist:		
keylist: yyyymdd, LOCID, FLT_TYPE, USER_CLASS, WEIGHT_CLASS, PHYS_CLASS, ETMS_EQPT, B_JET, R_JET, BUSAVIATION, IAC, ADG, TDG		
line: SELECT YYYMMDD, LOCID, FLT_TYPE, USER_CLASS, WEIGHT_CLASS, PHYS_CLASS, ETMS_EQPT, B_JET, R_JET, BUSAVIATION, AAC, ADG, TDG ,SUM(DEP_CNT)/1 AS FLT1 ,SUM(ARR_CNT)/1 AS FLT2 ,SUM(DEP_CNT+ARR_CNT)/1 AS FLT3 ,SUM(DEP_SEATS)/1 AS ST1 ,SUM(DEP_SEATS)/IFEQUAL(SUM(DEP_CNT),0,1,SUM(DEP_CNT)) AS ST2 ,SUM(ARR_SEATS)/1 AS ST3 ,SUM(ARR_SEATS)/IFEQUAL(SUM(ARR_CNT),0,1,SUM(ARR_CNT)) AS ST4 FROM LOCIDSIMS WHERE YYYMMDD IN (20241212,20241213,20241214,20241215,20241216,20241217,20241218) AND LOCID IN (' MC0') GROUP BY YYYMMDD, LOCID, FLT_TYPE, USER_CLASS, WEIGHT_CLASS, PHYS_CLASS, ETMS_EQPT, B_JET, R_JET, BUSAVIATION, AAC, ADG, TDG ORDER BY YYYMMDD, LOCID, FLT_TYPE, USER_CLASS, WEIGHT_CLASS, PHYS_CLASS, ETMS_EQPT, B_JET, R_JET, BUSAVIATION, AAC, ADG, TDG		
cmd: tot		
nopage: y		
nost: n		
defs:		
avgdays: 1		
oktosave: y		
locinput_param:		
locQuick_param:		

Step 1: Extract HTTP Headers

- Copy headers from browser's Network tab.
- Convert to a Python dictionary.

Example Header

```
1 headerDict = {  
2     "Accept": "text/html,...",  
3     "Content-Type": "application/x-www-form-urlencoded",  
4     "User-Agent": "Mozilla/5.0..."  
5 }
```

Step 2: Extract Cookies

- Copy cookies from browser.
- Split into a key-value dictionary.

Code

```
1 cookieDict = {  
2     "ASPSESSIONIDQSBCSBDD": "FFEFIKBANAJOPIGFDNHEA00I",  
3     "bm_sv": "49F07927D1A1B7D1AD7F21EAF73804A7..."  
4 }
```

Step 3: Prepare Payload

- Define query parameters (dates, airport ID, fields).
- Example: Fetch data for Orlando (MCO) from April 24–30, 2025.

Payload Fields

```
1 payloadDict = {  
2     "dlist": "20250424,20250425,...",  
3     "llist": "' MCO'",  
4     "keylist": "yyymmdd,LOCID,FLT_TYPE..."  
5 }
```

Step 4: Send POST Request

- Use `requests.post()` to fetch data.
- Include headers, cookies, and payload.

Code

```
1 url = "https://aspm.faa.gov/tfms/sys/tfms-server-x.asp"
2 response = requests.post(url, headers=headerDict,
3                           cookies=cookieDict, data=
    payloadDict)
```

Step 5: Save Results

- Check status code (e.g., 200 for success).
- Save HTML response to a file.

Code

```
1 with open("DelayReport_FAA.html", "w") as file1:  
2     file1.write(response.text)
```

- **Reverse Engineering:** Browser DevTools are essential
- **HTTP Mastery:** Understanding headers/cookies prevents blocking
- **Scalability:** Script can be extended for multiple airports

Try It Yourself!

- Run the script `hw3_automate_webRequest.ipynb`.
- Examine `DelayReport_FAA.html`.

Assignment 3: Web Automation

① Manually edit the payload

- Update the function in the script `hw3_automate_webRequest.ipynb` so that it accepts user-defined dates and airport code.
- Example input:
 - `dates = ['20250424', '20250425', '20250426', '20250430']`
 - `arpt_code = 'JFK'`
- Replace date and airport fields in the payload .
- Review and modify `payloadString` accordingly.

② Write a generalized function

Thank you!